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Is Cryptocurrency Wash Trading Strategic?

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Background and Context

- In early 2014, Mt.Gox transaction data dump was leaked
- Essentially, there is a database of bitcoin transactions from April 2011 to November 2013. This accounts for almost 80% of all bitcoin trades in the world at that time.
- The data includes transaction ID, amount, time, currency, transaction fees and other peripheral information

Literature Review

- Wash trading has been established in Aloosh and Li (2024) using this data. In particular, Gandal et al., 2018 found evidence of wash-trading in the MtGox dataset through two bots, while Aloosh and Li use statistical methods to show from a period of June 2011 to May 2013.
- Cong et al., 2023 find evidence of wash-trading using proprietary data from several exchanges through statistical methods
- There is no literature seeking to characterize the strategic nature of these wash trades.

New Findings

- Volume of wash trades increase when volume of non-wash trades is low, wash-trades go down when non-wash trades increase
- Wash trading sets off more trades across other platforms
- Wash trades should not be done when other assets are trading more. This is because BTC would itself be cannibalized by these other assets
- Wash traders capitalize on rumors and information asymmetry, such that when media interest is high, to do wash trades that would have a much more rapid effect but also tends to subside faster.
- Exogenous shock – Pot Day - Wash trade's effectiveness is limited by contemporaneous events which affect BTC's basic demand.

Table 2:

- Do crypto wash trades' volume or value matter?
- Volume of wash trades increase when volume of non-wash trades is low (replicating Aloosh and Li 2024)

Table 2. How do wash traders decide when to wash-trade? This panel presents coefficients from VAR examining lagged variables of liq_t, vol_t, $nonwash_t$, $total_t$ and their impact on $wash_t$. I use an arbitrary lag of up to $t-4$ The coefficients here do not refer directly to the variables in Table 1 but the weighted feature importance of each of them in the model. The higher the value, the more important the variable is. A placebo has been placed and as a guideline, any variable with importance lower than the placebo is not important as it shows that a randomly generated placebo has higher importance than it. Time period is from 26th June 2011 to 20th May 2013. The ranking of each variable in the corresponding model are shown in parentheses. Variable definitions are detailed in Table 1.

Feature Importance	(1)	(2)	(3)	(4)	(5)	(6)
	CART	Random Forest	AdaBoost	XGBoost	GRU	LTSM
liq_{t-1}	6.28e+07	0	0.0147	0.0165	2.95e-06	1.81e-08
	(7)	(11.5)	(16)	(14)	(8)	(9)
liq_{t-2}	8.46e+07	0	0.0271	0.0113	−1.11e-05	1.31e-08
	(4)	(11.5)	(9)	(16)	(14)	(10)
liq_{t-3}	7.57e+07	0	0.0207	0.0267	2.91e-06	4.46e-09
	(6)	(11.5)	(13)	(12)	(9)	(11)
liq_{t-4}	0	0	0.0217	0.0165	2.04e-06	−6.12e-10
	(15.5)	(11.5)	(12)	(15)	(11)	(12)
vol_{t-1}	1.45e+08	0	0.041	0.0971	4.89e-06	0.00016
	(2)	(11.5)	(7)	(4)	(5)	(1)
vol_{t-2}	0	0	0.022	0.034	4.4e-06	8.44e-05
	(15.5)	(11.5)	(11)	(10)	(6)	(2)
vol_{t-3}	2.5e+07	0.473	0.13	0.0325	3.06e-06	5.73e-05
	(11)	(1)	(4)	(11)	(7)	(3)
vol_{t-4}	4.65e+06	0	0.0205	0.0451	2.75e-06	2.86e-05
	(12)	(11.5)	(14)	(6)	(10)	(4)
$nonwash_{t-1}$	4.04e+07	0.0679	0.122	0.118	−2.23e-05	1.06e-07
	(8)	(3)	(5)	(3)	(15)	(8)
$nonwash_{t-2}$	2.88e+07	0.00613	0.028	0.0195	9.61e-06	2.94e-07
	(9)	(4)	(8)	(13)	(4)	(7)
$nonwash_{t-3}$	8.22e+07	0	0.0239	0.0349	−7.46e-06	9.67e-06
	(5)	(11.5)	(10)	(9)	(13)	(5)
$nonwash_{t-4}$	0	0	0.0161	0.0113	−2.06e-06	6.35e-06
	(15.5)	(11.5)	(15)	(17)	(12)	(6)
$total_{t-1}$	2.57e+07	0.00542	0.151	0.119	0.000139	−4.77e-07
	(10)	(5)	(2)	(2)	(1)	(13)
$total_{t-2}$	1.46e+08	0	0.156	0.24	7.35e-05	−9.04e-07
	(1)	(11.5)	(1)	(1)	(2)	(14)
$total_{t-3}$	1.32e+08	0.448	0.143	0.0918	−0.000156	−1.32e-05
	(3)	(2)	(3)	(5)	(16)	(17)
$total_{t-4}$	0	0	0.0136	0.0431	2.88e-05	−1.01e-05
	(15.5)	(11.5)	(17)	(8)	(3)	(16)
placebo	3.79e+05	0	0.0481	0.0432	−0.000254	−3.71e-06
	(13)	(11.5)	(6)	(7)	(17)	(15)
n	3.32e+04	3.32e+04	3.32e+04	3.32e+04	3.32e+04	3.32e+04

Table 3:

- Do crypto wash trades' volume or value matter?
- Through extended granger causality tests, wash-trades go down when non-wash trades increase

Table 3 - Panel A. How do wash traders decide when to wash-trade? This panel presents coefficients from a **Johansen test** examining variables of liq_t, vol_t, $nonwash_t$, $total_t$ and $wash_t$. Time period is from 26th June 2011 to 20th May 2013. Variable definitions are detailed in Table 1.

Hypothesized No. of Cointegrating Equations	Trace Statistic	0.95 Critical Value	0.95 Critical Value Max-Eigen	Pass/Fail (95% Significance)
At most 0	4.4e+04	60	30	Pass
At most 1	3e+04	40	24	Pass
At most 2	1.6e+04	24	18	Pass
At most 3	5.3e+03	12	11	Pass
At most 4	−2.1e+02	4.1	4.1	Fail

Panel B. How do wash traders decide when to wash-trade? This panel presents coefficients from a **Granger Causality Panel** examining variables of liq_t, vol_t, $nonwash_t$, $total_t$ and $wash_t$. Time period is from 26th June 2011 to 20th May 2013. Variable definitions are detailed in Table 1.

Independent Variable	Dependent Variable	Lag	F-test p-value	Result
$wash_t$	$nonwash_t$	1	0.018	Pass
$wash_t$	$nonwash_t$	2	5.6e-09	Pass
$wash_t$	$total_t$	1	3.4e-32	Pass
$wash_t$	$total_t$	2	1.5e-47	Pass
$wash_t$	liq_t	1	0.91	Fail
$wash_t$	liq_t	2	0.97	Fail
$wash_t$	vol_t	1	0.97	Fail
$wash_t$	vol_t	2	4.3e-107	Pass

Panel C. How do wash traders decide when to wash-trade? This panel presents coefficients from a VAR following the model in Table 2 with up to $t-4$ lags and its **Impulse Response Function** examining variables of liq_t, vol_t, $nonwash_t$, $total_t$ and $wash_t$. I use arbitrary forward time steps of up to $t+10$. Time period is from 26th June 2011 to 20th May 2013. Variable definitions are detailed in Table 1.

Lag	% IRF of $nonwash_t$ to $wash_t$	% IRF of $wash_t$ to $nonwash_t$	% IRF of $total_t$ to $wash_t$	% IRF of $wash_t$ to $total_t$
1	1.3e+04	-inf	6.7e+03	inf
2	1.7e+03	−9e+02	1e+03	93
3	−1.3e+05	2.8e+05	−2.1e+05	−1.2e+06
4	1.6e+03)	−4.3e+02	1.7e+03	2.5e+02
5	−1.3e+05	6.9e+05	−1.2e+05	−6.4e+05
6	2e+03	−60	2e+03	4.1e+02
7	−1.1e+05	−5.7e+06	−1.1e+05	−4.5e+05
8	2.3e+03	−59	1.8e+03	5.6e+02
9	−9.4e+04	−5.5e+06	−1.2e+05	−3.4e+05
10	2.3e+03	1.2e+02	2.3e+03	7.1e+02

Table 4:

- Can on-chain and cross-market data predict wash-trading?
- When wash trades are high, it sets off more transactions across other platforms and wallets, thus, periods of high wash trade in the MtGox exchange may mirror on-chain transaction activity more

Table 4 - Panel A. Do wash trades correlate with on-chain transactions? This panel presents coefficients from OLS regressions examining changes in non-wash trade volume to on-chain transaction volume. $nonwash_t$ and $onchain_t$ second-level data and accumulated to $t = 30$ minutes. Quartiles refer to whether the wash-trades are in the highest volume arranged to the lowest volume, with Quartile 4 being the highest trade volume and Quartile 1 being the lowest. Time period is from 16th September 2011 to 1st June 2012. p-values are shown in parentheses. Variable definitions are detailed in Table 1. *, **, *** represent significance at the 10%, 5% and 1% level.

	Quartile 1	Quartile 2	Quartile 3	Quartile 4
$nonwash_t$	0.447	−0.186	2.2***	−0.199
	0.475	0.68	5.27e-05	0.424
constant	2.4e+04***	2.16e+04***	2.3e+04***	1.95e+04***
	8.06e-120	7.12e-89	9.49e-46	9.07e-58
adj r^2	−0.000166	−0.000283	0.0052	−0.000122
n	2.96e+03	2.94e+03	2.94e+03	2.96e+03

Panel B. Do wash trades correlate with on-chain transactions? This panel presents coefficients from a two-step Engel-Granger cointegration examining changes in non-wash trade volume to on-chain transaction volume. $nonwash_t$ and $onchain_t$ seconds-level data and accumulated to $t = 30$ minutes. Quartiles refer to whether the wash-trades are in the highest volume arranged to the lowest volume, with Quartile 4 being the highest trade volume and Quartile 1 being the lowest. Time period is from 16th September 2011 to 1st June 2012.

Quartile	Cointegration p-value
Quartile 1	0.153
Quartile 2	0.0937
Quartile 3	0.0834
Quartile 4	0.0511*
n	2.94e+03

Panel C. Do wash trades correlate with other exchange transactions? This panel presents coefficients from OLS regressions examining changes in non-wash trade volume to market transaction volume. $nonwash_t$ and $market_t$ second-level data and accumulated to $t = 30$ minutes. Quartiles refer to whether the wash-trades are in the highest volume arranged to the lowest volume, with Quartile 4 being the highest trade volume and Quartile 1 being the lowest. Time period is from 26th June 2011 to 20th May 2013. p-values are shown in parentheses. Variable definitions are detailed in Table 1. *, **, *** represent significance at the 10%, 5% and 1% level.

	Quartile 1	Quartile 2	Quartile 3	Quartile 4
30-min-non-wash	0.0500***	0.1205***	0.1692***	0.2269***
	(0.0003)	(0.0000)	(0.0000)	(0.0000)
constant	5198.7580***	2666.5921***	1994.8933	−4762.9151**
	(0.0000)	(0.0000)	(0.1138)	(0.0209)
adj r^2	0.0685	0.3851	0.3397	0.5211
n	174	174	173	174

Table 5:

- Are wash trades motivated by other assets?
- This is evidence that wash trades are not random and does not take place when other markets are trading more actively
- It makes sense that wash trades should not be done when other assets are trading more. This is because BTC would itself be cannibalized by these other assets. Thus, wash trades can be taken to be strategic during this period.

Table 5 - Panel A. Do wash trades correlate with other asset movements? This panel presents coefficients from IRFs derived from VAR examining changes in wash trade volume $wash_t$ to other variables based on EURO-USD spot, Gold spot, SNP500 index and VIX. The prefixes represent each of the 4 assets respectively, and the suffixes - liq_t represents % change in liquidity, vol_t represents % change in realized volatility, $close_t$ represents % change in price, $tick_t$ represents % change in tick count and $volume_t$ represents % change in volume. The variables are seconds-level and accumulated to $t = 30$ minutes. Time period is from 26th June 2011 to 20th May 2013. Variable definitions are detailed in Table 1.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Lag	% IRF of $wash_t$ to $eur\text{-}vol_t$	% IRF of $wash_t$ to $eur\text{-}liq_t$	% IRF of $wash_t$ to $eur\text{-}tick_t$	% IRF of $wash_t$ to $eur\text{-}close_t$	% IRF of $wash_t$ to $gold\text{-}vol_t$	% IRF of $wash_t$ to $gold\text{-}liq_t$	% IRF of $wash_t$ to $gold\text{-}tick_t$	% IRF of $wash_t$ to $gold\text{-}close_t$
1	−199	−133	−260	−157	−212	−32.3	−171	−247
2	−105	−546	−3.88	−109	−123	7.88	126	−170
3	−254	−261	−199	3.11e+03	−94	−163	−247	−143
4	−47.8	−180	−30.2	−154	3.2e+03	2.92e+03	−173	−120
5	−20.2	−162	−6.83	−155	−157	−61.3	−106	−683
6	−221	−46.9	−113	−467	−73	−410	−1.02e+03	−112
7	−186	−159	−204	33.3	−896	−105	−338	−609
8	−119	64.8	−766	−217	−98.4	−227	−196	−361
9	4.98e+03	−276	155	−16.4	−8e+03	−60.4	−43.8	−144
10	−85.5	−185	−158	−226	−153	−120	−133	−241
n	3.33e+04	3.33e+04	3.33e+04	3.33e+04	3.33e+04	3.33e+04	3.33e+04	3.33e+04

Panel B. Do wash trades correlate with other asset movements? This panel presents coefficients from IRFs derived from VAR examining changes in wash trade volume $wash_t$ to other variables based on EURO-USD spot, Gold spot, SNP500 index and VIX. The prefixes represent each of the 4 assets respectively, and the suffixes - liq_t represents % change in liquidity, vol_t represents % change in realized volatility, $close_t$ represents % change in price, $tick_t$ represents % change in tick count and $volume_t$ represents % change in volume. The variables are seconds-level and accumulated to $t = 30$ minutes. Time period is from 26th June 2011 to 20th May 2013. Variable definitions are detailed in Table 1.

	(1)	(2)	(3)	(4)	(5)
Lag	% IRF of $wash_t$ to $snp\text{-}vol_t$	% IRF of $wash_t$ to $snp\text{-}liq_t$	% IRF of $wash_t$ to $snp\text{-}volume_t$	% IRF of $wash_t$ to $snp\text{-}close_t$	% IRF of $wash_t$ to $vix\text{-}close_t$
1	−345	−225	−217	−176	−428
2	−90.1	−72.8	−116	32.5	−105
3	50.3	−191	−207	−317	−4.77e+03
4	−58.1	−210	−114	−251	−167
5	−217	−14.4	−918	−135	−123
6	−754	−264	−21.4	−99.7	−24.9

Table 6:

- Does media attention affect wash-trades?
- This could mean that the wash trades affect non-wash trades much faster when there is more media intensity but the effects also subside much more quickly compared to when there is less media attention. Overall, this means that wash traders could be capitalizing on rumors and information asymmetry, such that when media interest is high, to do wash trades that would have a much more rapid effect but also tends to subside faster.

Table 6. Does media attention affect the effect of wash trades on market activity? This panel presents coefficients from IRFs derived from VAR examining changes in wash trade volume $wash_t$, non-wash trade volume $nonwash_t$, total trade volume $total_t$, liquidity liq_t, volatility vol_t. The variables are seconds-level and accumulated to $t =$ 30 minutes. They are then accumulated to weekly intervals. Weeks that do not have stationary data trends are dropped, thus, there are $n = 83$ points. *google* is a weekly datapoint and the median media popularity is based on whether the *google* for that week is higher or lower than the median. Columns (1)-(4) are for those below median media popularity, columns (5)-(8) are for those above median media popularity. Time period is from 26th June 2011 to 20th May 2013. Future time periods of $t + 1$ to $t + 10$ are shown and the sum is shown in the 11th row. Variable definitions are detailed in Table 1.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	% IRF of $total_t$ to $wash_t$	% IRF of $nonwash_t$ to $wash_t$	% IRF of $wash_t$ to $total_t$	% IRF of $wash_t$ to $nonwash_t$	% IRF of $total_t$ to $wash_t$	% IRF of $nonwash_t$ to $wash_t$	% IRF of $wash_t$ to $total_t$	% IRF of $wash_t$ to $nonwash_t$
	——Below median media popularity——				——Above median media popularity——			
1	−2.550e+05	−2.464e+05			1.268e+05	2.566e+05		
2	3.31e+03	4.16e+03	55.2	−9.18e+03	2.79e+04	−193	−647	946
3	3.324e+05	1.112e+06	1.792e+06	−2.96e+04	−1.736e+05	3.24e+04	2.183e+05	−1.43e+04
4	−148	156	259	−9.05e+03	−3.09e+04	64.9	−586	−1.21e+04
5	2.845e+05	3.006e+05	−1.009e+06	4.72e+03	−5.81e+04	−4.76e+04	1.440e+05	−8.36e+03
6	7.44e+03	8.65e+03	−90.8	−689	−692	488	−1.13e+03	2.22e+03
7	2.501e+05	3.74e+03	−4.23e+04	4.38e+03	6.38e+03	9.4e+04	3.39e+04	−8.66e+03
8	1.32e+03	1.55e+03	−129	−6.13e+03	343	−668	−600	−3.57e+03
9	−2.584e+05	−8.84e+04	−7.67e+04	2.45e+03	4.08e+04	2.4e+04	7.8e+03	−5.8e+03
10	889	235	−577	−2.91e+03	−968	4.44e+03	−1.19e+03	2.26e+03
sum	3.664e+05	1.096e+06	6.630e+05	−4.6e+04	−6.2e+04	3.634e+05	3.999e+05	−4.73e+04
n	83	83	83	83	83	83	83	83

Table 7

- Exogeneous shock - Pot Day, 20th April 2012
- This provides strong evidence that wash trade's effectiveness is limited by contemporaneous events which affect BTC's basic demand. When non wash trade increased, Columns (4) and (8) also show that wash trade generally decreased, which is in line with predictions that when the market returns to an active state, wash traders focus on trading with legitimate people compared to performing wash trades. The bigger -8.61e+03 in Column (4) compared to the smaller -6.94e+04 in Column (8) also shows that wash traders reduce their wash trading less in the aftermath of Pot Day when non wash trades increase, showing that they did more wash trades to hype up the market before Pot Day and decreased afterwards. This shows that wash trade behavior was strategic during this period, as they correlated with contemporary events and kept up wash trades more during popular times of BTC demand.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	%Δ SPY	%Δ SPY	%Δ SPY	%Δ SPY	%Δ SPY	%Δ VIX	%Δ VIX	%Δ VIX	%Δ VIX
Negative Facial	−0.005	−0.000	−0.004	−0.009***	−0.008***	−0.015	0.036	−0.029	−0.079***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)	(0.028)	(0.026)	(0.019)	(0.021)
Negative Facial i/r Conference Count	−0.000	−0.001***				−0.005***			
	(0.000)	(0.000)				(0.002)			
Negative Sentiment	0.003*	0.003*	0.003*	0.003*	0.003*	0.009	0.009	0.009	0.009
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.014)	(0.014)	(0.014)	(0.014)
Statement Related	−0.000**	−0.000**	−0.000**	−0.000**	−0.000**	−0.001	−0.000	−0.000	−0.001
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.002)	(0.002)	(0.002)	(0.002)
%Δ FDFD	−0.010	−0.006	−0.010	−0.012*	−0.008	0.066	0.122*	0.148*	0.018
	(0.008)	(0.008)	(0.008)	(0.007)	(0.007)	(0.068)	(0.070)	(0.078)	(0.067)
MPUUS_MPU	−0.000		−0.000***	−0.000***	−0.000	0.000	0.001***	0.001***	0.000
	(0.000)		(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Predrift SPY	−0.001	0.000	−0.002	−0.002	−0.000				
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)				
Negative * cfquart			−0.002**				−0.023***		
			(0.001)				(0.007)		
Negative * congre 30				0.001				0.038**	
				(0.002)				(0.019)	
Negative * congre 10					0.007*				−0.035
					(0.004)				(0.023)
Predrift VIX						−0.006**	−0.005**	−0.007***	−0.005**
						(0.003)	(0.002)	(0.002)	(0.003)
Chair FE	No	Yes	Yes	Yes	No	Yes	No	No	Yes
r2	0.012	0.028	0.039	0.037	0.014	0.054	0.017	0.013	0.050
N	1389.000	1389.000	1389.000	1389.000	1389.000	1359.000	1359.000	1359.000	1359.000